

Project design Phase

Proposed Solution

Date	29-06-2026
Project Name	Automated Network Request Management
Team ID	
Maximum Marks	

Here is the proposed architectural solution for the Automated Network Request Management (ANRM) project, detailing the core platform components, technical parameters, and their specific execution mechanics.

Proposed Architectural Solution

S.No.	Parameter	Description
1	User Intake Layer (Service Portal)	A dedicated self-service catalog item in ServiceNow that captures all required network parameters such as Source IP, Destination IP, Ports, Protocols, and Business Justification. Dynamic UI Policies and input validation ensure accurate and complete data entry.
2	Orchestration & Logic Engine (Flow Designer)	The central automation engine that triggers on request submission. It processes incoming variables, evaluates request risk levels, and executes conditional workflows such as auto-approval or escalation based on predefined business rules.

3	Integration Pipeline (Integration Hub)	Converts validated request data into structured payloads (JSON/XML) and securely communicates with external network systems through REST APIs, SSH, or orchestration spokes.
4	Network Infrastructure Execution Layer	The target infrastructure systems (e.g., firewalls, routers, IPAM tools) that receive ServiceNow commands and apply live network changes such as firewall rules, port openings, or routing updates.
5	Closed-Loop Verification Engine	Executes post-deployment connectivity or port validation checks to confirm successful implementation before marking the request as complete.
6	Configuration Track- Back (CMDB Update)	Automatically updates Configuration Items (CIs) and their dependencies in the CMDB after successful execution, ensuring infrastructure records remain accurate and up to date.
7	Compliance Tracking & Audit Trail	Maintains a complete and immutable log of all activities, including request details, approvals, integration payloads, API responses, and execution timestamps for compliance and auditing purposes.
8	Notification & Communication Layer	Sends automated real-time updates to stakeholders via email, Microsoft Teams, or Slack at key workflow stages such as submission, approval, deployment, and closure.